

KENAI, AK, USA

WMO: 702590

Lat: 60.580N

Lon: 151.239W

Elev: 28

StdP: 100.99

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26523

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-25.5	-32.8	0.2	-29.7	-28.4	0.3	-25.3	12.4	-2.2	11.5	-3.1	0.9	10	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	21.4	14.3	19.6	13.6	18.0	13.0	15.5	19.6	14.6	18.1	13.9	16.9	4.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.7	9.8	16.3	12.9	9.3	15.6	12.3	8.9	15.0	43.1	19.6	40.7	18.1	38.7	16.8	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	9.7	8.5	DB	-31.1	24.9	4.3	2.4	-34.2	26.6	-36.7	28.0	-39.1	29.4	-42.2	31.1	(4)
			WB	-31.1	16.9	4.2	1.9	-34.2	18.3	-36.6	19.4	-39.0	20.5	-42.0	21.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-9.0	-6.2	-4.2	2.5	7.7	11.4	13.5	12.8	9.1	2.5	-4.8	-7.5	(6)
(7)		DBStd	9.51	8.43	6.58	5.54	3.64	2.62	2.07	1.71	2.08	2.78	4.77	6.29	7.24	(7)
(8)		HDD10.0	3067	588	454	441	224	81	8	0	3	47	232	445	543	(8)
(9)		HDD18.3	5830	847	688	699	474	331	208	150	171	277	489	695	802	(9)
(10)		CDD10.0	279	0	0	0	0	8	50	109	91	20	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	18	0	0	0	0	1	6	7	4	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.4	3.6	3.8	3.6	3.8	3.8	3.5	3.1	3.2	3.3	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	482	27	25	22	19	25	28	44	67	80	67	40	39	(15)
(16)		PrecMax	688	77	70	81	46	70	62	91	137	180	187	176	101	(16)
(17)		PrecMin	280	2	5	1	0	2	2	7	14	18	18	0	5	(17)
(18)		PrecStd	100	20	14	20	14	17	16	22	32	41	36	34	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	5.7	8.0	14.6	22.1	23.6	23.9	19.0	14.0	7.8	5.4	(19)	
(20)			MCWB	3.0	3.1	3.9	7.6	12.0	14.5	15.8	16.1	12.6	10.0	5.1	3.3	(20)
(21)		2%	DB	3.7	3.9	5.7	11.6	18.1	20.3	20.8	16.8	11.0	5.8	3.7	(21)	
(22)			MCWB	2.0	2.0	2.6	6.0	10.2	13.1	14.7	14.8	11.6	8.1	4.0	2.1	(22)
(23)		5%	DB	2.4	2.7	4.0	10.0	15.3	18.0	19.1	19.0	15.3	9.8	4.3	2.5	(23)
(24)			MCWB	1.1	1.2	1.3	5.3	8.7	12.0	14.1	13.7	11.1	7.1	3.1	1.2	(24)
(25)		10%	DB	1.3	1.6	2.8	8.5	13.3	16.2	17.7	17.5	13.9	8.7	3.1	1.3	(25)
(26)			MCWB	0.2	0.4	0.7	4.5	7.8	11.1	13.6	13.3	10.7	6.5	1.9	0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.5	4.5	8.7	13.2	15.3	16.8	17.2	13.6	10.6	5.6	3.7	(27)
(28)			MCDWB	5.0	5.0	7.2	12.8	20.6	22.0	21.8	22.2	17.1	12.8	7.1	5.0	(28)
(29)		2%	WB	2.3	2.3	3.0	6.8	10.7	13.7	15.4	15.6	12.5	8.8	4.3	2.4	(29)
(30)			MCDWB	3.4	3.6	5.1	10.7	16.6	18.9	19.5	19.5	15.3	10.4	5.4	3.4	(30)
(31)		5%	WB	1.3	1.3	1.9	5.6	9.4	12.6	14.7	14.7	11.8	7.8	3.2	1.4	(31)
(32)			MCDWB	2.3	2.5	3.6	9.2	14.2	16.7	18.1	17.8	14.1	9.3	4.2	2.3	(32)
(33)		10%	WB	0.2	0.5	1.0	4.8	8.5	11.7	14.0	13.8	11.1	6.8	2.0	0.5	(33)
(34)			MCDWB	1.2	1.5	2.5	8.0	12.5	15.3	16.9	16.5	13.3	8.3	3.0	1.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	9.5	8.8	9.5	8.4	7.4	8.4	8.3	7.6	7.5	7.4	(35)
(36)			MCDBR	5.4	6.2	8.4	11.0	14.4	13.2	11.3	12.4	10.8	7.7	5.4	5.4	(36)
(37)			MCWBR	4.6	5.0	6.0	6.7	7.8	7.0	6.2	7.1	6.8	5.3	4.3	4.6	(37)
(38)		5% WB	MCDBR	5.4	5.7	7.2	9.8	12.2	11.1	9.4	10.0	8.5	6.5	5.1	5.4	(38)
(39)			MCWBR	4.7	4.7	5.3	6.4	7.1	6.2	5.6	6.2	5.7	4.8	4.2	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.285	0.303	0.343	0.367	0.358	0.366	0.351	0.323	0.300	0.251	0.225	(40)	
(41)		taud	2.168	2.296	2.330	2.330	2.340	2.451	2.460	2.504	2.562	2.494	2.174	1.982	(41)	
(42)		Ebn at Noon	531	716	828	854	856	873	855	842	811	706	494	344	(42)	
(43)		Edh at Noon	38	64	85	103	110	101	98	87	69	53	37	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.29	0.96	2.51	4.05	5.19	5.43	4.78	3.80	2.46	1.17	0.42	0.15	(44)	
(45)		RadStd	0.04	0.13	0.29	0.40	0.43	0.39	0.47	0.45	0.35	0.12	0.07	0.02	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.82	+3.51	+3.74	N/A	N/A	N/A	-271	-321	N/A	N/A	(46)
(47)	Regional (22 neighbors)	+0.68	+2.15	+1.89	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page